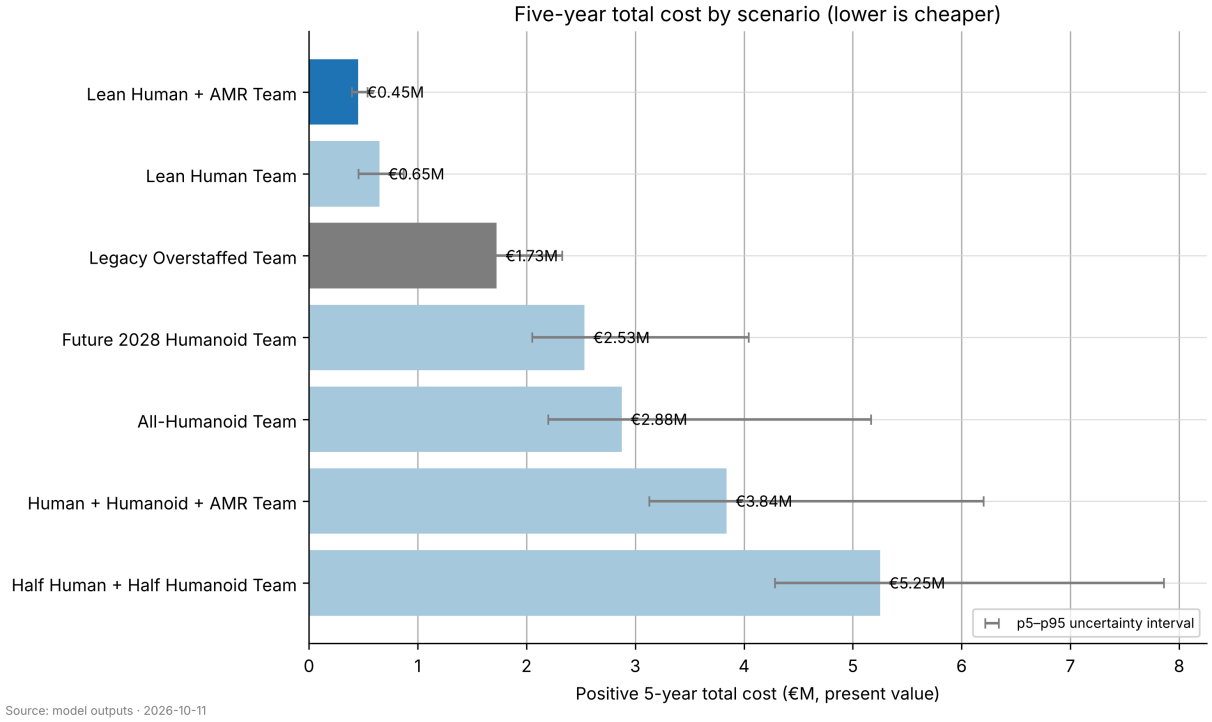
Warehouse humanoid TCO — decision exhibits

Deploy Lean Human + AMR Team now; defer humanoid procurement

Lean Human + AMR Team costs €453K over five years and is the lowest-cost feasible scenario (1 human + 3 AMRs from the demand frontier). The least-cost humanoid-inclusive case, Future 2028 Humanoid Team, costs €2.533M and remains above the human baseline.



Source: model outputs · 2026-10-11

The old eight-human baseline is not the decision baseline

The retained baseline is a transparent legacy reference. Fair comparison starts with a demand-sized crew under the $\rho \leq 0.85$ constraint.

Robot ranking: Future 2028 Humanoid Team leads humanoids, but does not win overall

```
'Humanoid leader: Future 2028 Humanoid Team (-2,532,753 EUR NPV); lean hybrid: -452,706 EUR.'
```

Integration makes hybrid-AMR worse, not better

Human + Humanoid + AMR Team carries the €200K system-integration cost retained under F-222 and is costlier than Future 2028 Humanoid Team despite its AMR operating design.

Capacity is feasible; cost is the discriminating result

At modeled demand, all published scenarios are demand-bound. Capacity-ceiling analysis establishes feasibility; it does not create robot savings.

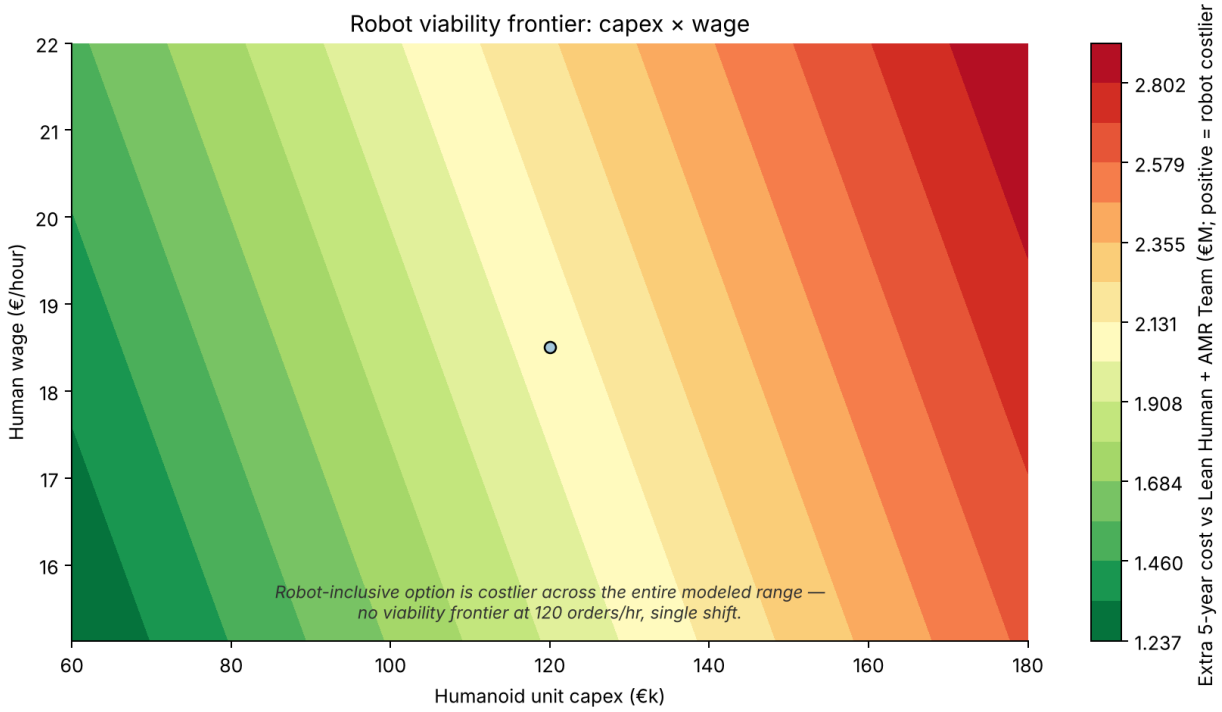
Transfer from household demonstrations is the largest validity risk

The source episodes are not warehouse tasks. A 0.50–0.90 transfer factor conservatively translates capability, but local telemetry should replace it.

Wage and capex do not reverse the published decision grid

```
'No tie inside €60k–€180k capex with other assumptions fixed.'
```

See chart 07 for the capex × wage frontier and chart 08 for capex × transfer.



Sensitivity is paired and transparent

Monte Carlo uses common random numbers across scenarios. Sobol-style values are coarse-grid variance proxies, explicitly not full interaction estimates.

Decision gate: evidence required before capital commitment

Measure warehouse task cycle times, robot availability, integration scope, and unfilled-vacancy cost in a local pilot; then rerun the model.

Audit trail and challenge path

The decision memo, data lineage, raw report JSON, assumptions register, and objection FAQ are all versioned in this repository.